Neural Network Optimizers

A Comparative Study of SGD, RMSProp and Adam Using the CIFAR-10 Dataset

1. Introduction

Modifying the parameters of a neural network model while training is for a correct mapping of input data to output. Weights of the model are updated according to the gradients of a loss function through optimization algorithms. The optimizer is one of the choices that can impact efficiency of training, stability of learning and resulting performance of the model.

Various optimizers utilize differing strategies for updating weights on neural networks. Some techniques use straightforward gradient updates; others resume the learning rate to improve speed. Grasping how this optimization algorithms operate in real life is useful to construct machine learning systems.

This tutorial investigates the behaviour of three commonly used neural network optimizers:

- Stochastic Gradient Descent (SGD)
- RMSProp
- Adam (Adaptive Moment Estimation)

The goal of this tutorial is to demonstrate how these optimizers influence neural network training behaviour using a practical experiment. A neural network model is trained on the CIFAR-10 image classification dataset, and the performance of each optimizer is analysed using several visualizations and evaluation metrics.

2. Dataset Overview

The experiments of this tutorial use the CIFAR-10 dataset, a dataset that has become a benchmark dataset for computer vision and deep learning research.

CIFAR-10 is made up of 60,000 colour images belonging 10 categories. All images are 32 × 32 pixels. These images consist of various objects like animals, vehicles, and aeroplanes.

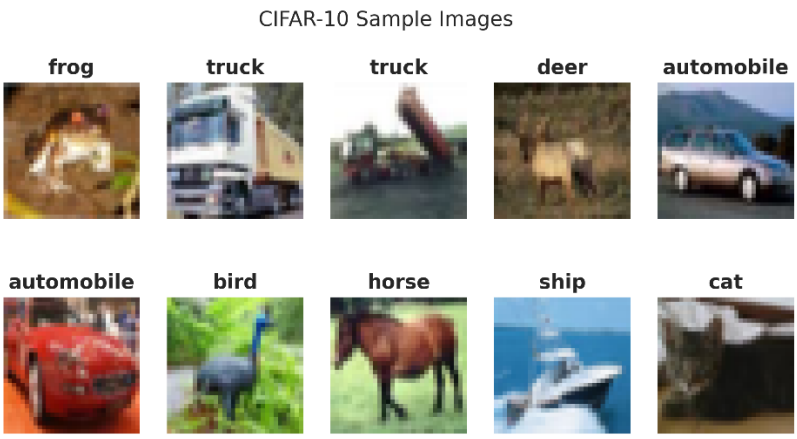
The dataset is divided into:

- 50,000 training images
- 10,000 testing images

The categories included in the dataset are:
airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Because the images are small and contain diverse visual features, CIFAR-10 provides a challenging classification problem. This makes the dataset useful for studying how optimization algorithms affect neural network learning behaviour.

Before training the model, the image pixel values were normalized to the range 0–1 to ensure stable gradient updates during training.



As shown in Figure, a few images from the CIFAR-10 dataset. Every image represents one of the ten categories of objects contained in the dataset. The image portrays the various types of objects and visual patterns in the dataset.

Since the images are of low resolution, it becomes difficult to differentiate between two similar categories, cat and dog. The CIFAR-10 dataset can be a useful framework for studying training behaviour of neural networks and performances of optimisers.

3. Neural Network Model

To ensure a fair comparison between optimizers, the same neural network architecture was used for all experiments. The model used in this tutorial is a **Multilayer Perceptron (MLP)** consisting of fully connected layers.

The architecture includes the following components:
Flatten Layer - Converts the input image into a one-dimensional feature vector.
Hidden Layer 1 – 256 neurons with ReLU activation
Hidden Layer 2 - 128 neurons with ReLU activation
Output Layer – 10 neurons with softmax activation representing the probability of each class

4. Optimization Algorithms

Three optimization algorithms were evaluated in this tutorial.

Stochastic Gradient Descent (SGD):
SGD is one of the most fundamental optimization methods used in machine learning. Instead of computing gradients using the entire dataset, SGD updates model weights using smaller mini-batches of data. This reduces computational cost and allows the model to learn continuously during training.
However, SGD may converge slowly if the learning rate is not properly tuned.

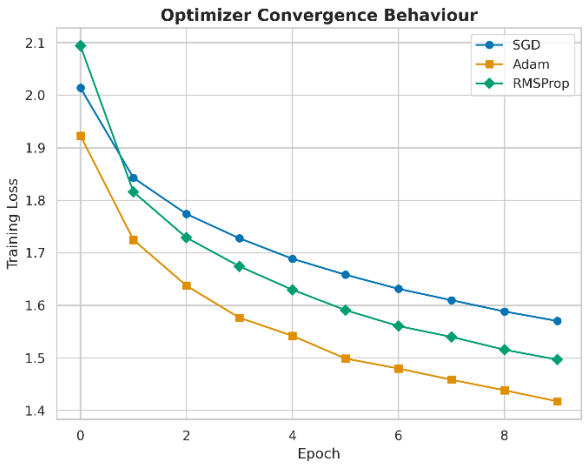
RMSProp:
RMSProp adapts the learning rate of each parameter in gradient descent. It keeps a moving average of the squared gradients and modifies the updates. This method helps stabilize training when the gradients are quite different.

Adam Optimizer:
Adam combines the advantages of momentum optimization and adaptive learning rate techniques. It computes running averages of both gradients and squared gradients, allowing the optimizer to adapt learning

rates during training. Adam is widely used in modern deep learning because it often converges faster than traditional optimization methods.

5. Training Behaviour Analysis

To understand how each optimizer behaves during neural network training, the loss values were monitored across training epochs.

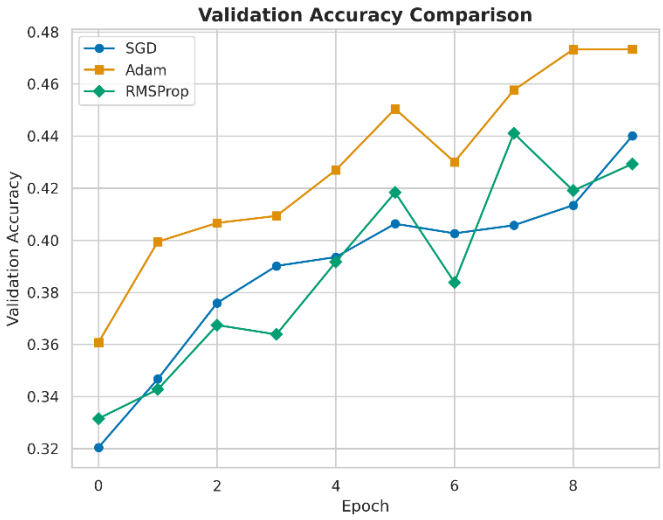


This figure illustrates the training loss curves for each optimizer. Training loss represents the difference between the predicted outputs of the model and the true labels.

A decreasing loss indicates that the neural network is learning effectively. The plot shows that adaptive optimizers such as Adam reduce the loss more rapidly during early training stages. RMSProp also demonstrates stable convergence, while SGD typically reduces the loss more gradually.

6. Validation Accuracy Analysis

While training loss indicates how well the model fits the training data, validation accuracy provides insight into how well the model generalizes to unseen data.

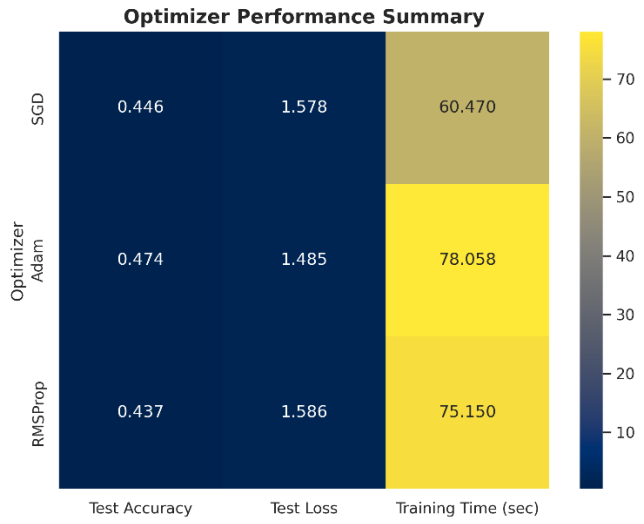


Above figure shows how validation accuracy evolves during training for each optimizer. Higher validation accuracy indicates better predictive performance on unseen data.

Adam generally achieves higher validation accuracy earlier in the training process, suggesting that it learns useful patterns more efficiently. RMSProp demonstrates steady improvements in accuracy, while SGD improves more slowly.

These trends illustrate how optimizer choice can influence the model's ability to generalize beyond the training dataset.

7. Optimizer Performance Comparison

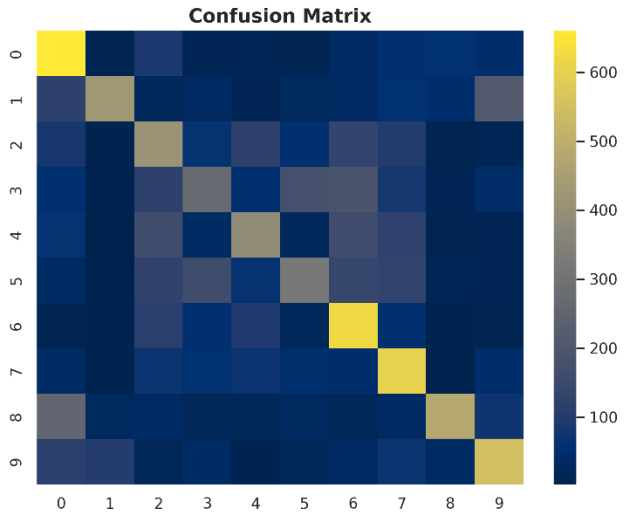


The test accuracy, loss values, and training time, among other key metrics for the optimizers, are summarized in this figure.

The heatmap enables readers to easily compare the performance of each optimizer across multiple criteria. The Model Performance is Better if Accuracy Values are High. The time taken for training gives extra insight on inefficiency. This helps in spotting the efficient optimizer which is good in training and has good prediction with less training time.

8. Model Evaluation

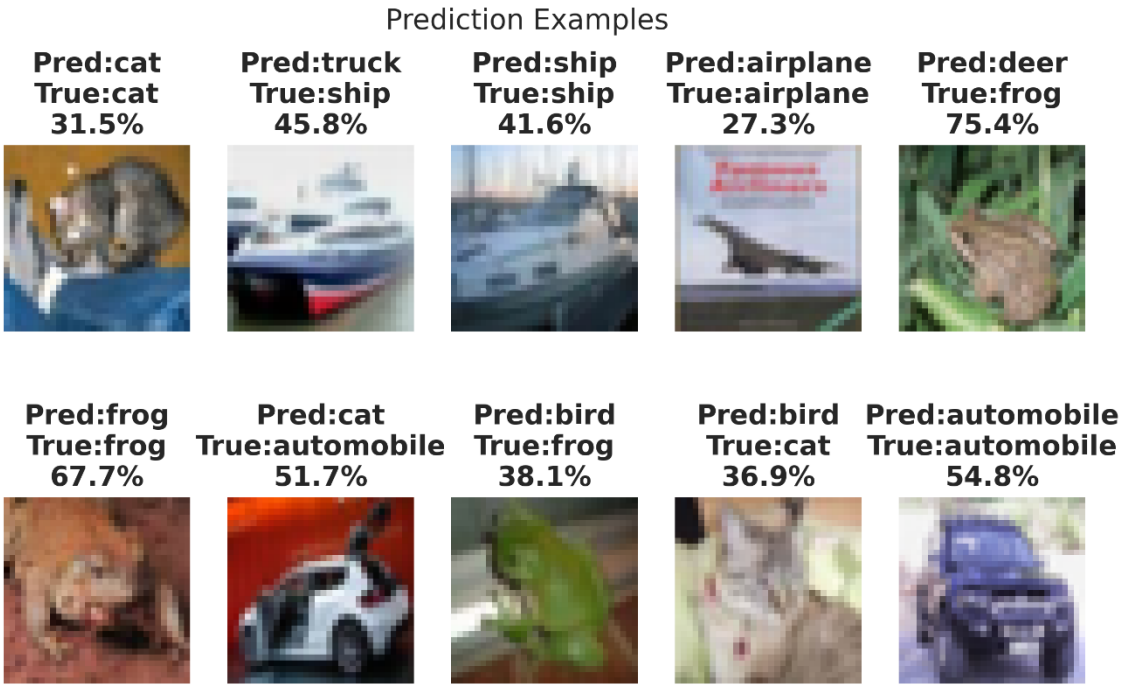
After training the models, the best performing optimizer was selected and evaluated using a confusion matrix.



The confusion matrix shows how accurately the trained model classifies various object classes. Classes that are true are represented by the rows whereas the predicted classes are represented by the columns.

The matrix shows correct predictions on its diagonal. Off-diagonal values imply mistakes. Through the analysis of these values, we can identify which object categories present more challenges to the model.

9. Example Model Predictions



As shown the above figure , several test images along with predicted class labels, true labels and confidence scores are displayed. These examples give us a sense of what the model will do in practice.

The correct predictions reveal the model's capacity to recognize objects even with changes. Misclassified examples show the challenge of differentiating objects having similar appearances. Illustrative examples help relate quantitative evaluation metrics to actual model behaviour.

10. Discussion

Findings from the experiment show that choice of optimizer significantly affects training behaviour of neural network.

Adaptive optimizers such as Adam generally converged with improved stability compared to vanilla gradient descent. In comparison to SGD, RMSProp maintains stable training behaviour and requires the use of considerably less epochs.

The results also show that we must choose the right optimizer when training neural networks. Different optimization algorithms perform differently depending on the dataset, the model architecture and the training conditions.

11. Conclusions

The key findings of this tutorial include:

- Adam demonstrated the fastest convergence during neural network training.
- RMSProp provided stable training behaviour with competitive accuracy.
- SGD required more training iterations to achieve comparable performance.
- Optimizer selection significantly influences neural network learning behaviour.

Understanding these differences helps machine learning practitioners choose appropriate optimization strategies for their models.

12. References

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